PROJECT

7.1

SET UP A WEB APP IN THE CLOUD

DEVOPS SERIES

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Objective

This project aims to demonstrate how to deploy a Java-based web application in the cloud using AWS services.

It involves setting up secure user permissions, launching a virtual server, connecting through VSCode, installing required development tools, and deploying the application remotely.

Services

AWS IAM - Controls secure access with user permissions.

Amazon EC2 - Hosts the web app on a cloud-based VM.

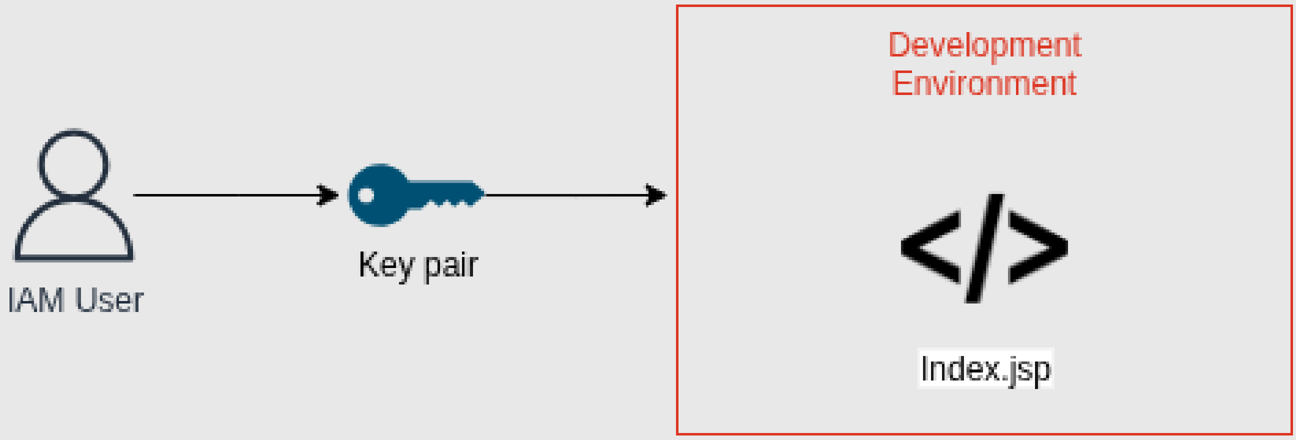
Amazon Corretto 8 - Provides Java runtime for the app.

Apache Maven - Builds and manages Java projects.

VSCode - Used for writing and editing code.

Remote - SSH - Connects VSCode to EC2 for remote development.

Architecture Diagram



An IAM user generates SSH keys to access the EC2 instance. Using Remote-SSH, VSCode connects securely to the cloud-based development environment. Inside EC2, tools like Corretto and Maven are used to build and run the application (e.g., Index.jsp), enabling live cloud-based development.

Steps to Implement

1. Create an IAM user:
Set up a new IAM user with programmatic access and attach permissions for EC2 and SSH access to ensure secure resource management.

2. Launch an EC2 Instance:
Spin up an EC2 instance using Amazon Linux 2 or Ubuntu, configure security groups to allow SSH, and download the key pair for later connection.

3. Install VSCode and Remote - SSH:
Install Visual Studio Code along with the Remote - SSH extension to enable coding directly on your EC2 instance from your local machine.

4. Connect to EC2 from VSCode:
Use the .pem key and Remote - SSH extension to establish a secure connection between VSCode and your EC2 instance for remote development.

5. Install Corretto 8 and Maven on EC2:
Install Amazon Corretto 8 as the Java runtime and Apache Maven for building and managing your Java application on the EC2 instance.

6. Create and deploy the application:
Set up a Maven-based Java project, write your application code, and build it using Maven directly on the EC2 instance through VSCode.

Story Time

IAM (The Gatekeeper of Permissions):
IAM is your mighty gatekeeper who hands out magical badges (permissions) to chosen warriors (users). Only those with the right badge can enter the EC2 realm and command cloud resources.

EC2 (The Virtual Cloud Fortress):
EC2 is your stronghold in the sky, a virtual castle where your application will live. You launch it with a special key (key pair) and guard it with shield spells (security groups).

VSCode (The Developer's Magic Scroll):
VSCode is your enchanted scroll that lets you write powerful spells (code). When combined with the right tools, it becomes your gateway to building anything in your cloud kingdom.

Remote - SSH (The Secret Portal):
This extension is your secret portal to teleport into your cloud castle (EC2). With your magic key (.pem file), you step directly into the server from your scroll (VSCode) and start casting code spells.

Amazon Corretto 8 & Maven (The Java Spellbook & Forge):
Corretto is your Java spellbook, filled with ancient coding powers, while Maven is the forge where you build, compile, and manage your project potions and artifacts.

Web App Creation (The Grand Build):
You craft your web app using Java, shape it with Maven, and bring it to life right inside your cloud castle. With every build, you add a new rune to your magical masterpiece.

BY
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THANK YOU!